The Experiential Reinforcement Learning Paradigm: Structural Deconstruction, Constraint Integration, and the Latent Cognitive Bottleneck

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Abstract

This paper provides a comprehensive structural and theoretical analysis of the Experiential Reinforcement Learning (ERL) paradigm as an advancement over Reinforcement Learning with Verifiable Rewards (RLVR) for training autonomous large language model agents. The paper deconstructs the mathematical formulations underlying ERL's experience-reflection-consolidation loop, evaluates its integration with Inverse Constraint Learning (ICL) for safety-critical deployment, and benchmarks its empirical performance against contemporaneous frameworks including Branch-and-Rethink Reward Models and Self-Aware Guided Efficient Reasoning. Following this analysis, a novel theoretical critique is introduced identifying "Reasoning Collapse" as a fundamental vulnerability in ERL's internalization mechanism. To resolve this vulnerability, a new optimization framework—Differentiable Reflection-State Preservation (DRSP)—is proposed, enforcing latent space alignment during the consolidation phase to preserve cognitive geometry without incurring inference-time computational overhead.

Keywords: experiential reinforcement learning, large language models, inverse constraint learning, differentiable reflection-state preservation, agentic reasoning, reasoning collapse

Introduction to Agentic Large Language Models and Optimization Trajectories

The deployment of large language models as autonomous, decision-making agents has necessitated a profound paradigm shift in computational optimization strategies. Historically, Supervised Fine-Tuning (SFT) provided these models with static imitation capabilities, allowing for robust pattern reproduction derived from human-annotated datasets. While SFT establishes the foundational syntax and semantic priors necessary for interaction, it is fundamentally an open-loop system, offering no intrinsic mechanism for behavioral revision or dynamic adaptation once the model is deployed in an interactive environment. As agentic requirements grew increasingly complex, necessitating multi-step planning, tool utilization, and environmental interaction, Reinforcement Learning from Human Feedback (RLHF) and subsequently Reinforcement Learning with Verifiable Rewards (RLVR) emerged as the dominant frameworks.

RLVR allows models to learn interactively by optimizing against scalar feedback mechanisms, enabling test-time scaling and the capacity to explore complex solution spaces through extended chains of thought. However, RLVR inherently relies on undirected trial-and-error methodologies. In environments characterized by sparse, delayed, and partial information, models are forced to implicitly infer how observed failures map to necessary behavioral corrections, a process that is notoriously unstable, computationally expensive, and sample-inefficient. The fundamental limitation lies in the reduction of complex, multi-dimensional environmental interactions to single scalar endpoints. Multi-step reasoning tasks amplify minor early-stage errors, obfuscating credit assignment across the trajectory and forcing the policy into a cycle of back-and-forth exploration without durable, structural behavioral correction. Furthermore, recent analyses of large reasoning models indicate that extending the length of reasoning chains does not monotonically correlate with correctness; in many instances, length inflation introduces substantial redundancy, impairing computational efficiency and occasionally deteriorating accuracy.

In response to these critical limitations, a novel training paradigm termed Experiential Reinforcement Learning (ERL) has been introduced, shifting the objective from purely outcome-driven scalar optimization to structured, semantic learning from experience. Drawing inspiration from cognitive theories of experiential learning, ERL embeds an explicit experience-reflection-consolidation loop directly into the reinforcement learning trajectory, transforming raw scalar feedback into intermediate, actionable reasoning.

This paper conducts an exhaustive deconstruction of the ERL framework, its underlying mathematical formulations, its integration with Inverse Constraint Learning (ICL) for safety-critical reliability, and its comparative standing against contemporaneous methods such as

Branch-and-Rethink Reward Models. Following this empirical deconstruction, a novel theoretical critique is introduced, demonstrating how the current ERL internalization mechanism inadvertently triggers Reasoning Collapse in out-of-distribution scenarios. To resolve this critical vulnerability, a new theoretical optimization framework—Differentiable Reflection-State Preservation (DRSP)—is proposed to maintain latent cognitive structures without incurring the autoregressive inference penalties of traditional reflection.

The Theoretical Limitations of Reinforcement Learning with Verifiable Rewards

To fully appreciate the architectural innovations of ERL, one must first rigorously analyze the mathematical and structural limitations of the baseline RLVR framework. In standard RLVR, the interaction between the language model agent and the environment is formalized as a Markov Decision Process (MDP), defined by the tuple containing the state space, action space, transition dynamics, and a scalar reward function. The objective is to find a policy parameterized by a set of neural network weights that maximizes the expected cumulative reward over a specified time horizon.

Recent advancements in large reasoning models have predominantly utilized Group Relative Policy Optimization (GRPO) to optimize this objective. GRPO simplifies the traditional Proximal Policy Optimization (PPO) algorithm by eliminating the need for a separate value network (critic), instead estimating the baseline by sampling a group of multiple responses for the same input prompt and normalizing the rewards within that specific group. The policy is updated by maximizing a surrogate objective that incorporates a clipping mechanism to prevent destructively large policy updates, alongside a Kullback-Leibler divergence penalty to ensure the updated policy remains proximate to the reference policy, thereby preserving linguistic coherence and preventing reward hacking.

While GRPO is highly efficient for well-defined tasks with dense reward signals, such as deterministic mathematical proofs, its efficacy degrades precipitously in sparse-reward control environments and multi-turn agentic tasks. In an environment like Sokoban, where the agent must execute a precise sequence of discrete movements to achieve a goal state, the reward is strictly terminal. The agent receives a reward only upon successful completion of the entire puzzle; all intermediate states yield a reward of zero. Under the GRPO framework, if a group of sampled trajectories all fail to reach the terminal goal state, the variance of the rewards within the group is zero, rendering the advantage estimation mathematically meaningless and resulting in vanishing gradients.

Consequently, the policy must rely on random autoregressive sampling to stumble upon the correct sequence of actions before any meaningful gradient update can occur. This undirected exploration is computationally prohibitive in environments with vast state spaces. Furthermore, when the agent does receive a negative terminal reward, the scalar nature of the feedback provides no semantic information regarding the specific point of failure within the trajectory. The model cannot distinguish whether the failure was caused by a tactical error in the final step or a fundamental strategic miscalculation in the first step. This obfuscation of credit assignment prevents the model from engaging in durable behavioral correction, relegating it to a process of blind trial-and-error that scales poorly to highly complex, long-horizon tasks.

Structural Deconstruction of the Experiential Reinforcement Learning Paradigm

ERL directly addresses the credit assignment and undirected exploration bottlenecks of standard RLVR by restructuring the learning process around semantic experience rather than scalar optimization. The core innovation is the embedding of an explicit experience-reflection-consolidation loop inside the trajectory generation phase, allowing the model to perform targeted behavioral correction before policy consolidation.

Mathematical Formulation of the Experience-Reflection-Consolidation Loop

The ERL algorithm fundamentally alters the MDP interaction by inserting a structured verbalization phase between the observation of an environmental outcome and the subsequent policy update. This algorithmic loop operates through a rigorously defined sequence of generative and evaluative phases.

In the first phase, denoted as the Initial Experience, the base language model policy receives an input task query and autoregressively generates an initial attempt trajectory. The agent executes this trajectory within the given environment and receives corresponding environmental feedback, which may include explicit error messages, state changes, or a terminal scalar reward. Rather than immediately utilizing this scalar reward to compute an advantage estimate for a policy update, the algorithm evaluates the reward against a predefined success threshold.

If the initial reward falls below this threshold, indicating a suboptimal performance or outright failure, the algorithm triggers the Conditional Self-Reflection phase. The model utilizes the concatenated history of the task input, its own initial attempt, and the resulting environmental feedback to generate a structured verbal critique. This reflection is mathematically conditioned on a cross-episode reflection memory buffer, which serves as a repository of previously successful heuristics. The generation of this reflection forces the language model to externalize its latent error attribution, explicitly diagnosing the failure modes of the initial attempt and proposing strategic revisions in natural language.

Following the generation of the self-reflection, the algorithm enters the Refined Experimentation phase. The model is prompted to produce a second, refined attempt at the task, conditioned on the original input and the newly generated reflection critique. The environment evaluates this revised attempt, yielding a secondary set of environmental feedback and a final scalar reward. The reflection is subsequently assigned the reward value of this second attempt, creating a direct reinforcement signal that encourages the generation of critiques that causally lead to higher downstream performance.

The final phase of the online loop involves Memory Consolidation. If the refined second attempt yields a substantial improvement that surpasses the success threshold, the successful reflection string is appended to the cross-episode memory buffer. This accumulation of knowledge allows the agent to persist successful corrective priors across different training episodes, stabilizing future reflection generation and encouraging the reuse of previously discovered effective strategies.

Optimization and the Internalization Mechanism

To ensure both the generative quality of the reflection and the ultimate behavioral adaptation of the policy, ERL leverages a dual-phase optimization strategy. The primary reinforcement learning objective utilizes a policy gradient estimator to update the network weights over the entire interacting sequence, encompassing the first attempt, the reflection, and the second attempt. The advantage estimate is computed from the rewards associated with the respective phases, aligning the model's behavior with the environmental objectives.

However, the most critical architectural component of the ERL framework is its approach to deployment efficiency, referred to as the Internalization Mechanism. While inference-time reflection significantly improves task performance by allowing the model to engage in System 2 deliberative thinking, generating the intermediate reasoning tokens for the reflection and the subsequent second attempt is computationally prohibitive for real-time, low-latency applications. ERL resolves this via selective self-distillation.

Given a training episode where the second attempt was successful, the framework utilizes supervised fine-tuning to force the model to imitate the improved behavior conditioned solely on the original input query, effectively marginalizing out the reflection tokens and the initial failed attempt. The distillation loss minimizes the negative log-likelihood of the successful trajectory given only the starting state. This process forces the neural network parameters to internalize the corrective logic articulated in the reflection, theoretically enabling the model to reproduce the optimized behavior zero-shot at inference time without requiring the explicit reflection scaffold or the associated computational overhead.

Empirical Validation: Sparse-Reward Control and Agentic Reasoning Benchmarks

The empirical efficacy of the ERL paradigm has been rigorously evaluated across a diverse spectrum of sparse-reward control environments and complex agentic reasoning benchmarks, utilizing mid-scale open-weights models such as Qwen3-4B-Instruct and Olmo-3-7B-Instruct. The experimental results provide compelling evidence that structured experiential revision consistently outperforms standard RLVR in both final reward attainment and overall sample efficiency.

Table 1

Comparative Performance of RLVR Baseline vs. ERL Across Benchmark Environments

Evaluation Environment	Primary Cognitive Challenge	RLVR Baseline Reward (Qwen)	ERL Reward (Qwen)	Relative Performance Gain
Sokoban	Long-horizon planning, recovery from compounding errors	0.06	0.87	+81.0%
FrozenLake	Unknown dynamics inference, sparse terminal rewards	0.86	0.94	+8.0%
HotpotQA	Tool-use, multi-hop retrieval, answer synthesis	0.45	0.56	+11.0%

The most profound empirical demonstration of ERL's superiority is observed in the Sokoban environment, where the Qwen3-4B-Instruct model improved from a baseline reward of 0.06 to an extraordinary 0.87, representing an absolute gain of 81 percentage points. Sokoban requires precise, long-horizon spatial planning, where a single incorrect block push can render the puzzle mathematically unsolvable, leading to compounding errors. Standard RLVR fails catastrophically in this domain because the scalar terminal reward cannot isolate which specific move in a fifty-step sequence doomed the trajectory. ERL allows the model to explicitly deduce environmental rules through language—for instance, noting that pushing a block into a corner creates an unrecoverable state—and incorporate this deduction into the subsequent attempt, effectively bypassing the need for millions of undirected exploratory episodes.

In the FrozenLake environment, which demands that the agent infer abstract symbol semantics, action consequences, and terminal conditions purely through interaction under sparse rewards, the model improved from 0.86 to 0.94. Notably, the evaluation design deliberately withheld explicit game rules, forcing the model to acquire task understanding exclusively through trial-and-error. This demonstrates that the experience-reflection-consolidation loop enables the model to analyze failures, revise spatial strategies, and internalize corrective behavior within a single episode, accelerating convergence compared to traditional methods.

The HotpotQA benchmark, adapted into an agentic multi-hop question-answering task requiring iterative tool-assisted retrieval, showed more moderate but highly reliable gains, improving from 0.45 to 0.56. The smaller relative gain is attributable to the underlying task structure; HotpotQA presents a more homogeneous interaction pattern centered on repeated tool invocation with denser evaluation feedback. Because the baseline RL algorithm already receives relatively informative gradients in this regime, the marginal benefit of structured experiential revision is reduced compared to the grid-based control environments. This contrast suggests that ERL yields the greatest mathematical advantage in environments where learning requires substantial semantic reasoning about unknown dynamics and long-horizon consequences, rather than tasks that primarily optimize over a stable, short-horizon interaction loop.

Evaluating the Memory Paradox: Ablation Dynamics and Contextual Reuse

To rigorously isolate the mechanistic contributions of individual algorithmic components within the ERL framework, extensive ablation studies were conducted, removing specific architectural elements while maintaining the baseline reinforcement learning hyperparameters. These studies reveal critical insights into the dynamics of agentic self-improvement and expose potential vulnerabilities regarding memory accumulation.

The ablation of the structured reflection component—wherein the model retains the two-attempt interaction structure but the explicit self-reflection generation is replaced with a generic text prompt encouraging improvement—results in the most severe degradation of policy quality. On the FrozenLake environment utilizing the Qwen3-4B-Instruct architecture, the removal of reflection caused the final reward to plummet from 0.94 to 0.60, falling substantially below even the baseline RLVR performance. A similar catastrophic drop from 0.87 to 0.59 was observed in Sokoban. This empirically validates that the detailed semantic content of the self-generated critique, rather than the mere computational opportunity to retry the task, is the primary driver of optimization and behavioral correction.

Conversely, the ablation of the cross-episode reflection memory reveals a highly nuanced, environment-dependent dynamic. The no-memory variant disables the persistent storage of successful reflections, meaning corrective signals remain localized to individual trajectories and cannot be retrieved as context for future episodes. In relatively stable environments like FrozenLake, the removal of memory slowed the rate of convergence and slightly reduced the final reward from 0.94 to 0.86, suggesting that memory facilitates the propagation of useful heuristics across the training distribution.

However, in the highly complex and stochastic setting of Sokoban utilizing the Olmo3-7B-Instruct model, the no-memory variant actually outperformed the full ERL implementation, achieving a final reward of 0.24 compared to the baseline 0.20. This anomaly exposes a critical theoretical phenomenon termed the "Memory Paradox." When a model's intrinsic self-reflective capacity is limited, or when the environment is exceptionally unforgiving, the agent is prone to generating inaccurate, spurious, or hallucinated reflections early in the training process. If these flawed heuristics happen to coincide with a stochastic success, they are committed to the persistent cross-episode memory buffer. Consequently, future reflections are conditioned on this poisoned prior, causing the agent to lock into sub-optimal behavioral manifolds based on hallucinated environmental dynamics. In such adversarial topologies, disabling cross-episode memory acts as a necessary regularizer, preventing the accumulation and propagation of erroneous logical priors.

Comparative Methodologies: Branch-and-Rethink Reward Models and SAGE

The principles underlying ERL are not occurring in a vacuum; parallel research trajectories in reward modeling and inference optimization are exploring similar mechanisms of deliberate, multi-step reasoning. An analysis of these contemporaneous frameworks provides essential context for understanding the broader shift toward reasoning-centric architectures.

Branch-and-Rethink Reasoning Reward Model

Traditional reward models utilized in alignment pipelines compress numerous complex quality dimensions—such as factuality, safety, reasoning coherence, and stylistic adherence—into a single scalar value in a single forward pass. This monolithic design induces a phenomenon termed "judgment diffusion," where the model's attention is spread indiscriminately across all evaluation criteria, resulting in a diluted focus and shallow analysis that frequently misses subtle but highly consequential errors.

To rectify this, the Branch-and-Rethink Reasoning Reward Model (BR-RM) transfers the "think-twice" principle from solver models directly to the reward modeling architecture. Operating as a two-turn generative reward model, BR-RM closely mirrors the structural logic of ERL. In the first turn, Adaptive Branching, the model explicitly avoids holistic grading and instead selects a small subset of instance-critical dimensions from a universal set, generating a concise, evidence-seeking hypothesis regarding potential weaknesses in the target response. This operates analogously to the diagnostic reflection phase in ERL.

In the second turn, Branch-Conditioned Rethinking, the model executes a targeted re-read of the response, specifically testing the hypotheses generated in the first turn and scrutinizing only the flagged dimensions. The model is trained using GRPO over these structured two-turn traces, utilizing a strict binary outcome reward combined with heavy negative penalties for format deviations. Empirical evaluations demonstrate that this multi-turn, focused analysis allows an 8-billion parameter BR-RM model to outperform single-turn, 32-billion parameter baselines across challenging benchmarks like RewardBench and RM-Bench, proving that allocating test-time compute to structured rethinking is vastly more efficient than merely scaling parameter counts.

Self-Aware Guided Efficient Reasoning

While ERL forces the model to engage in explicit reflection to overcome the limitations of standard autoregressive generation, research into Self-Aware Guided Efficient Reasoning (SAGE) suggests that large reasoning models actually possess an implicit understanding of when a reasoning chain is complete and correct, but this capability is obscured by standard temperature sampling or greedy decoding.

Standard Pass@1 sampling often forces models into severe "length inflation," generating hundreds of redundant tokens even after the correct answer has been internally derived, simply because the probability distribution flattens. SAGE introduces a novel sampling paradigm that expands the exploration width during inference, retaining multiple candidate reasoning branches based on cumulative log-probability scores. The critical finding of SAGE is that when a model generates the specific termination token, it assigns extraordinarily high confidence to this token only when the preceding reasoning chain is concise and highly accurate.

By integrating this metric into group-based reinforcement learning (SAGE-RL), the framework forces the policy to learn these concise, high-confidence reasoning patterns without requiring explicit verbal reflection. SAGE-RL achieves up to a 44.1% reduction in generated tokens while simultaneously improving accuracy across mathematical benchmarks, highlighting an alternative to ERL's explicit verbalization: directly unlocking and reinforcing the latent efficient pathways already present within the model's probability manifold.

Integrating Inverse Constraint Learning into the ERL Framework

While the foundational ERL framework exhibits exceptional capabilities in reward maximization and task success, the deployment of autonomous agents in safety-critical domains necessitates strict, mathematically verifiable adherence to operational boundaries. Standard RL objectives inherently incentivize reward hacking, wherein the agent exploits environmental loopholes that achieve high scalar rewards but violate human safety expectations or system parameters.

To address the critical requirement of agent reliability, the theoretical architecture of ERL has been extended to incorporate Inverse Constraint Learning (ICL). This integrated framework focuses on the autonomous discovery of latent safety constraints from the agent's own failures and reflections, moving beyond simple task performance to ensure robust operation under distribution shifts.

Reflection as a Diagnostic Constraint Signal

The core hypothesis of the ERL-ICL extension is that reflection-generated corrections contain high-fidelity semantic signals that can be systematically utilized to infer latent safety constraints that were not explicitly programmed into the environment's base reward function. In this advanced framework, the role of reflection undergoes a crucial paradigm shift: instead of utilizing the reflection solely as an intermediate mapping for reward enhancement, the framework parses the natural language reflection to identify explicit rule violations.

When an agent fails a task, the generated reflection frequently highlights specific failure modes that transcend mere sub-optimality and cross the threshold into unsafety. The framework targets three specific categories of violations commonly articulated in agentic reflections:

1. Incorrect Reasoning Steps: Logical inconsistencies or flawed deductions that lead to computationally invalid or physically dangerous actions.
2. Unsafe Tool Usage: The invocation of external APIs, bash terminals, or physical effectors in a manner that violates system parameters, bypasses access controls, or risks catastrophic system failure.
3. Hallucinated Information: The integration of fabricated facts, nonexistent variables, or unfounded assertions into the decision-making chain, leading to unpredictable downstream consequences.

These reflection-derived signals act as the foundational dataset for Inverse Constraint Learning. The framework utilizes these signals to train a differentiable Constraint Predictor. The parameters of the constraint model are optimized via a supervised binary cross-entropy objective. State-action pairs that the textual reflection identifies as constraint violations are assigned a hard label, whereas actions deemed safe and productive are labeled accordingly. Consequently, the neural network learns to approximate a continuous probability surface representing the boundary of safe operations within the state-action space of the environment.

Lagrangian Relaxation and the Reflection-Constrained Policy Improvement Operator

With the constraint predictor established, the underlying RL formulation is upgraded from a standard MDP to a Constrained Markov Decision Process (CMDP). The optimization objective fundamentally shifts from pure reward maximization to constrained maximization. The agent must maximize expected cumulative rewards while ensuring that the expected cost—as predicted by the learned constraint model—remains below a predefined threshold of zero.

To solve this CMDP practically and continuously during training, the framework employs Lagrangian relaxation. A dual variable, known as the Lagrangian multiplier, is introduced to control the penalty for violating the learned constraint model. The unconstrained saddle-point objective optimized by the policy gradient algorithm is formalized as: $L(\theta, \lambda) = E[R] - \lambda \cdot E[Z_\varphi(s, a)]$.

To operationalize this within the experiential loop, the framework introduces a specialized Reflection-Constrained Policy Improvement Operator. During the refinement phase, the standard reflection generates an improved trajectory aimed at maximizing the base reward. The operator takes this proposed trajectory and applies a projection function, mapping the proposed behavior onto the nearest point within the constraint-satisfying policy space derived from the predictions of the constraint predictor.

This algorithmic integration ensures that the final behavior distilled back into the base policy during the internalization phase is not only optimized for the scalar reward but is strictly bounded by the safety rules autonomously discovered through the agent's prior reflections. The resulting metric of success is Agent Reliability, formally defined as the probability of constraint satisfaction when the agent is subjected to an out-of-distribution state distribution.

The Theoretical Critique: Self-Distillation, Template Collapse, and Out-of-Distribution Brittleness

While the integration of ERL and ICL presents a mathematically elegant and empirically robust mechanism for extracting and utilizing safety constraints during training, a rigorous theoretical analysis reveals a fundamental structural flaw in the framework's terminal phase: the Internalization Mechanism.

The ERL framework relies entirely on supervised self-distillation to bypass the prohibitive computational cost of generating autoregressive reflection tokens during real-time deployment. By optimizing the distillation loss, the framework effectively performs behavioral cloning on its own successful, reflection-guided outputs. However, this specific objective function is mathematically blind to the causal mechanism that produced the improved output. The refined attempt was generated conditioned on the semantic reasoning, error attribution, and constraint awareness embedded in the reflection. The distillation phase attempts to force the marginal distribution to match this output exactly. In doing so, it collapses the multi-step cognitive reasoning graph into a direct, shallow stimulus-response reflex.

This architectural shortcut leads directly to a phenomenon recently identified in advanced computational linguistics as "Reasoning Collapse" or "Template Collapse." Recent analyses demonstrate that when a large language model is distilled to output final answers without the intermediate "thinking tokens," it loses the informational peaks necessary to navigate complex logical state spaces. The model learns the surface-level heuristic of the safe, high-reward action but discards the generalized physical, logical, or safety laws articulated in the reflection. It becomes a stochastic parrot of its own prior reasoning.

This collapse poses a catastrophic threat to the framework's stated goal: achieving reliability under distribution shift. Consider a safety-critical scenario where the parameters of the environment undergo a subtle semantic or physical shift. An agent utilizing the full experiential loop (with active reflection) would fail once, reflect on the failure, identify the new constraint signal, and succeed on the retry by utilizing its System 2 deliberative reasoning capacity.

However, a deployed agent relying on the internalized, distilled policy has had its reflective capacity marginalized out. It executes the behavior that was deemed safe in the training distribution as a rigid, unyielding heuristic. Because the distilled policy lacks the explicit cognitive representation of the constraint, it cannot dynamically adjust. The constraint predictor may correctly identify internally that the action is now unsafe, but if the primary policy has been distilled into a System 1 reflex, the capacity for adaptive intervention is completely lost. Thus, while ICL improves training-time constraint discovery, the reliance on token-level self-distillation mathematically ensures out-of-distribution brittleness.

A Novel Optimization Framework: Differentiable Reflection-State Preservation

To resolve the contradiction between the computational necessity of internalization (avoiding inference-time token generation) and the catastrophic out-of-distribution brittleness caused by Reasoning Collapse, this paper proposes a strict, novel theoretical framework: Differentiable Reflection-State Preservation (DRSP).

The central thesis of this proposed framework is that token-level behavioral distillation fundamentally degrades out-of-distribution constraint adherence by marginalizing the causal reasoning graph. Robust internalization requires the projection of the reflection's latent representation directly into the base policy's continuous vector space, acting as a profound structural regularizer rather than a simple behavioral target.

The objective of DRSP is to force the model's internal hidden states to route information and activate attention heads as if the reflection were physically present in the context window, without actually requiring the autoregressive generation of reflection tokens during deployment. Let $h_\theta(x)$ represent the final hidden state representation of the input x before the language modeling head, produced by the deployed policy operating without reflection. Let $h_\theta(x, \Delta)$

represent the final hidden state representation of the input concatenated with the optimal reflection Δ, produced by the teacher policy during the experiential consolidation phase.

Instead of merely minimizing the cross-entropy over the discrete tokens of the refined attempt, DRSP introduces a latent alignment loss using a divergence-based objective. During the internalization phase, the total loss is reformulated as a composite function: $L\_internalize(\theta) = \alpha \cdot L\_distill(\theta) + \beta \cdot L\_DRSP(\theta)$. The DRSP loss is defined using the Kullback-Leibler divergence (or an alternative metric such as Cosine Similarity for dense high-dimensional vectors) between the latent states. A lightweight, trainable multi-layer perceptron (MLP) adapter maps the base representation into the dimensionality of the reflection space, and the stop-gradient operator is utilized to stabilize the teacher representations and prevent representation collapse during joint training.

Theoretical Implications of DRSP on Cognitive Topologies

The implementation of DRSP provides three profound theoretical and practical advantages over standard ERL self-distillation:

1. Preservation of Cognitive Geometry: By aligning the latent states through Kullback-Leibler divergence, the framework forces the attention matrices of the model processing only the prompt to activate the exact same semantic pathways that would have been activated had it explicitly "thought" about the constraints articulated in the reflection. This mathematically preserves the internal logical structure—the underlying reasoning topology—rather than just cloning the output tokens. The model retains the complexity of System 2 thinking while executing with System 1 speed.
2. Robustness to Distribution Shifts: When the deployed agent faces a state sampled from an out-of-distribution environment, the latent representation has been explicitly trained to

embed the implicit safety boundaries and constraint topologies discovered by the constraint model during training. Consequently, the model's internal logits will naturally skew away from constraint violations even in highly novel scenarios, preventing the brittle, heuristic-driven failure modes characteristic of standard self-distillation.

3. Zero Inference Penalty Maintenance: At deployment, the model requires absolutely no additional autoregressive decoding steps to generate reflections. The forward pass processes the input and outputs the safe, optimized action directly. This maintains the exact computational efficiency and low-latency execution of the original ERL framework, while successfully inheriting the safety and robustness of explicit, methodical reflection.

Conclusion

The evolution from scalar-driven RLVR to ERL represents a profound maturation in the optimization of autonomous language agents. By introducing an explicit experience-reflection-consolidation loop, ERL allows models to transcend blind trial-and-error, instead diagnosing their own failures and converting sparse environmental feedback into actionable, structured behavioral revisions. The extraordinary empirical gains demonstrated in complex, multi-step environments such as Sokoban rigorously validate the necessity of integrating intermediate semantic cognitive steps into the reinforcement learning pipeline.

Furthermore, the theoretical integration of ERL with Inverse Constraint Learning establishes a rigorous, mathematically sound mechanism for the autonomous discovery of latent safety rules. By leveraging reflection as a diagnostic constraint signal and applying Lagrangian relaxation to solve the resulting Constrained Markov Decision Process, the framework successfully transitions agents from pure reward maximization to safe, boundary-aware optimization.

However, the current framework's ultimate reliance on autoregressive self-distillation for deployment internalization presents a critical vulnerability. The reduction of a multi-step causal reasoning graph into a reflexive behavioral output induces Reasoning Collapse, severely compromising the out-of-distribution reliability the constraint learning framework seeks to guarantee.

The proposed Differentiable Reflection-State Preservation framework provides a definitive theoretical resolution to this bottleneck. By enforcing latent space alignment during the consolidation phase via Kullback-Leibler divergence mapping, DRSP ensures that the architectural geometry and safety boundaries of the reflection are permanently and structurally embedded within the model's weights. This evolution will allow future language agents to operate with the unparalleled computational speed of reflexive heuristics while maintaining the robust, constraint-aware adaptability of deliberate, experiential reasoning.

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